

GEiPRS Vignette

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Introduction

GEiPRS is a package that is used to fit the group lasso or sparse group lasso on big genomics data. We assume the data are stored in `.pgen/.pvar/.psam` format by the PLINK library. In addition, it also requires the `.pvar.zst` file which can be generated using `zstd`. For example, `zstd yourdata.pvar` will generate `yourdata.pvar.zst`.

The potential training/validation split can be specified with a separate column in the phenotype file.

The most essential parameters in the core function `geiprs` include:

- **genotype.pfile**: the PLINK 2.0 pgen file that contains genotype. We assume the existence of `genotype.pfile.{pgen,pvar.zst,psam}`.
- **phenotype.file**: the path of the file that contains the phenotype values and can be read as a table.
- **phenotype**: the name of the phenotype. Must be the same as the corresponding column name in the phenotype file.
- **env**: the name of the environment variable. Must be the same as the corresponding column name in the phenotype file. This is required to run the group lasso or sparse group lasso.
- **tau**: the sparse group lasso mixing parameters. If $\tau = 0$, it is group lasso penalty; if $\tau = 1$, it is lasso penalty. Only environment `!= NULL` can consider this parameter.
- **covariates**: a character vector containing the names of the covariates included in the lasso fitting, whose coefficients will not be penalized. The names must exist in the column names of the phenotype file.
- **family**: the type of the phenotype: "gaussian". Currently only "gaussian" family is supported.
- **alpha**: the elastic-net mixing parameter, where the penalty is defined as $\alpha \cdot \|\beta\|_1 + (1 - \alpha) \cdot \|\beta\|_2^2/2$. **alpha** = 1 corresponds to the lasso penalty, while **alpha** = 0 corresponds to the ridge penalty.
- **split.col**: the column name in the phenotype file that specifies the membership of individuals to the training or the validation set. The individuals marked as "train" and "val" will be treated as the training and validation set, respectively. When specified, the model performance is evaluated on both the training and the validation sets.
- **mem**: Memory (MB) available for the program. It tells PLINK 2.0 the amount of memory it can harness for the computation. IMPORTANT if using a job scheduler.

Some additional important parameters for model building include:

- **nlambda**: the number of lambda values on the solution path.
- **lambda.min.ratio**: the ratio of the minimum lambda considered versus the maximum lambda that makes all penalized coefficients zero.
- **p.factor**: a named vector of separate penalty factors applied to each coefficient. This is a number that multiplies lambda to allow different shrinkage. If not provided, default is 1 for all variables. Otherwise should be complete and positive for all variables.

The other parameters can be specified in a config list object, such as `missing.rate`, `MAF.thresh`, `nCores`, `num.snps.batch` (batch size M of the BASIL algorithm), `save` (whether to save intermediate results), `results.dir`, `prevIter` (when starting from the middle), `use.glmnetPlus` and `glmnet.thresh` (convergence threshold). More details can be seen in the function documentation. In particular, If we want to recover

results and continue the procedure from a previous job, we should have `save = TRUE` and specify `prevIter` with the index of the last successful (and saved) iteration.

GEiPRS depends on two other programs **plink2** and **zstdcat**. If they are not already on the system search path, it is important to specify their locations in the `configs` object and pass it to `geiprs`.

```
configs <- list(
  # results.dir = "PATH/TO/SAVE/DIR", # needed when saving intermediate results
  # save = TRUE, # save intermediate results per iteration (default FALSE)
  # nCores = 16, # number of cores available (default 1)
  # niter = 100, # max number of iterations (default 50)
  # prevIter = 15, # if we want to start from some iteration saved in results.dir
  # use.glmnetPlus = TRUE, # recommended for faster computation
  # early.stopping = FALSE, # whether to stop based on validation performance (default TRUE)
  # verbose = TRUE, # print intermediate messages (default FALSE)
  plink2.path = "plink2", # path to plink2 program
  zstdcat.path = "zstdcat" # path to zstdcat program
)
# check if the provided paths are valid
for (name in names(configs)) {
  tryCatch(system(paste(configs[[name]], "-h"), ignore.stdout = T),
    condition = function(e) cat("Please add", configs[[name]], "to PATH, or modify the path in the config",
  )
}
```

Example

We utilize the example data to illustrate the application of the GEiPRS package in constructing the genotype main effect-based polygenic risk score (PRS_G) and the genotype-environment interaction effect-based polygenic risk score (PRS_{GEI}) using the sparse group lasso method.

```
library(geiprs)
```

For the following script, when `split.col` is specified, validation metric will automatically be computed. Moreover, by default, early stopping is adopted to stop the process based on the validation metric (the extent controlled by `stopping.lag` in `configs` object).

```
genotype.pfile <- file.path(system.file("extdata", package = "geiprs"), "example.trainingANDvalidation")
phenotype.file <- system.file("extdata", "example.phe", package = "geiprs")
phenotype <- "resid"
env <- "envir"

fit_geiprs <- geiprs(
  genotype.pfile = genotype.pfile,
  phenotype.file = phenotype.file,
  phenotype = phenotype,
  env = env,
  tau = 0.5,
  split.col = "split", # split column name in phenotype.file with train/val/test labels
  # mem = 128000, # amount of memory available (MB), recommended
  configs = configs,
  family = "gaussian"
) # we hide the intermediate messages
```

The intercept and coefficients can be extracted by `fit_geiprs$a0` and `fit_geiprs$beta`. It also saves the evaluation metric, which by default is R^2 for the Gaussian family.

```

fit_geiprs$metric.train %>% na.omit()
#> [1] 0.008461921 0.008862384 0.009194857 0.009470881 0.009700041 0.009890293
#> [7] 0.010048245 0.011032154 0.012302843 0.014015410 0.015008158 0.016645767
#> [13] 0.018190636 0.020204719 0.021965781 0.023775334 0.025919799 0.028847561
#> [19] 0.031890599 0.035460488 0.041692037 0.047256194 0.053562951 0.061154862
#> [25] 0.070468005 0.081997302 0.094645266 0.108257568 0.123175322
#> attr(,"na.action")
#> [1] 30 31 32 33 34 35 36 37 38 39 40 41 42 43 44 45 46 47 48
#> [20] 49 50 51 52 53 54 55 56 57 58 59 60 61 62 63 64 65 66 67
#> [39] 68 69 70 71 72 73 74 75 76 77 78 79 80 81 82 83 84 85 86
#> [58] 87 88 89 90 91 92 93 94 95 96 97 98 99 100
#> attr(,"class")
#> [1] "omit"
fit_geiprs$metric.val %>% na.omit()
#> [1] 0.006373289 0.006608089 0.006789217 0.006927011 0.007029947 0.007104961
#> [7] 0.007157722 0.007783825 0.008608358 0.009705663 0.010013090 0.010520377
#> [13] 0.011428409 0.012675864 0.013706372 0.014588136 0.015603121 0.017032251
#> [19] 0.018411689 0.020174339 0.023717423 0.025382932 0.026471078 0.027475441
#> [25] 0.028078560 0.028914028 0.029815378 0.028715901 0.027943687
#> attr(,"na.action")
#> [1] 30 31 32 33 34 35 36 37 38 39 40 41 42 43 44 45 46 47 48
#> [20] 49 50 51 52 53 54 55 56 57 58 59 60 61 62 63 64 65 66 67
#> [39] 68 69 70 71 72 73 74 75 76 77 78 79 80 81 82 83 84 85 86
#> [58] 87 88 89 90 91 92 93 94 95 96 97 98 99 100
#> attr(,"class")
#> [1] "omit"

```

Next, we can get the optimal idx based on the validation metric. For the testing set, we can use the `predict_geiprs` function to predict the PRS_G and PRS_{GEI} with the fitted object `fit_geiprs`, based on the optimal idx.

```

optimal_idx <- which.max(fit_geiprs$metric.val)

test.genotype.pfile <- file.path(system.file("extdata", package = "geiprs"), "example.testing")

pred_geiprs <- predict_geiprs(
  fit = fit_geiprs,
  new_genotype_file = test.genotype.pfile,
  new_phenotype_file = phenotype.file,
  phenotype = phenotype,
  env = env,
  split_col = "split",
  split_name = c("test"),
  idx = optimal_idx,
  configs = configs)
#> Warning in checkMissingPhenoWarning(phe.master, phe.no.missing.IDs): We detected missing values for .

names(pred_geiprs)
#> [1] "prediction"          "prs_g"              "prs_ge"
#> [4] "response"           "env_val"            "metric"
#> [7] "metric.prs"         "metric.prs.logpval"

```

Now we can get predicted PRS_G and PRS_{GEI} .

```

cat("Predicted PRS_G:", "\n")
#> Predicted PRS_G:
pred_prs_g <- pred_geiprs[['prs_g']][['test']]
print(head(pred_prs_g))
#>                                     s26
#> 0-unrelatedBritish11 0.076431798
#> 0-unrelatedBritish13 0.065903629
#> 0-unrelatedBritish16 0.064349067
#> 0-unrelatedBritish20 0.005729977
#> 0-unrelatedBritish28 0.042642733
#> 0-unrelatedBritish29 0.007233464

cat("Predicted PRS_GEI:", "\n")
#> Predicted PRS_GEI:
pred_prs_gei <- pred_geiprs[['prs_ge']][['test']]
print(head(pred_prs_gei))
#>                                     s26
#> 0-unrelatedBritish11 0.4401397
#> 0-unrelatedBritish13 -0.1572999
#> 0-unrelatedBritish16 0.1790730
#> 0-unrelatedBritish20 0.1202063
#> 0-unrelatedBritish28 0.2163110
#> 0-unrelatedBritish29 -0.1257624

```

R^2 explained by *envir*, PRS_G , and $envir * PRS_{GEI}$ is stored in `pred_geiprs[['metric.prs']][['test']]`.

```

pred_geiprs[['metric.prs']][['test']]
#> [1] 0.04388405

```

For the regression model: $phenotype \sim envir + PRS_G + envir * PRS_{GEI}$, the $-\log_{10}$ (P value of the $envir * PRS_{GEI}$) is stored in `pred_geiprs[['metric.prs.logpval']][['test']]`.

```

pred_geiprs[['metric.prs.logpval']][['test']]
#> [1] 66.84141

```